

Revised
Generation Interconnection
System Impact Study Report

For

PJM Generation Interconnection Request
Queue Position AC2-103

Beaver – Davis Bessie 345kV

May 2020
Revision 2

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

Revision from April 2020 System Impact Study Report

The results from a recent MISO Affected System Impact Study retool are included in the Network Impacts section. Results show that there are contingency MISO MTEP upgrades that AC2-103 will need in-service or will need MISO to perform an interim study before going commercial.

Revision from November 2018 System Impact Study Report

The System Impact Study has been retooled due to behind the meter units in ATSI being dispatched incorrectly in the AC2 2020 model. These behind the meter units were contributing to the overload observed on the Beaver-Black River 138 kV line identified in the original AC2-103 System Impact Study report. This overload no longer exists through the AC2 Queue. AC2-103 no longer needs to be subject to interim deliverability studies to check the status of the planned baseline upgrades on this line.

General

Firelands Wind, LLC, the Interconnection Customer (IC), has proposed a wind generating facility located in Erie County, Ohio. The installed facilities will have a total capability of 297.66 MW with 38.69 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project is December 31, 2021. **This study does not imply a ATSI commitment to this in-service date.**

Point of Interconnection

AC2-103 will interconnect with the American Transmission Systems Inc.(ATSI) transmission system along the Beaver – Davis Besse 345kV line.

The point of interconnection is on the Beaver-Davis Bessie 345kV line, approximately 28.1miles from Beaver and 31.8miles from Davis Besse 345kV substations at the same point of interconnection as the Emerson Creek Wind Farm Beaver-Davis Besse 345kV (X1-027A) POI.

Cost Summary

The AC2-103 project will be responsible for the following costs:

Description	Cost	Tax (if applicable)	Total Cost
Attachment Facilities	\$ 0	\$ 0	\$ 0
Direct Connection Network Upgrades	\$ 10,863,300	\$ 1,393,900	\$ 12,257,200
Non Direct Connection Network Upgrades	\$ 757,400	\$ 97,300	\$ 854,700
Allocation for New System Upgrades	\$ 0	\$ 0	\$ 0
Contribution for Previously Identified Upgrades	\$ 0	\$ 0	\$ 0
Total Costs	\$ 11,620,700	\$ 1,491,200	\$ 13,111,900

General Information

Interconnection Customer ("IC"): Firelands Wind LLC Queue Position: AC2-103

Interconnected

Transmission Owner ("TO"): American Transmission Systems, Incorporated ("ATSI")

Impacted TO(s)
(if applicable):

American Transmission Systems, Incorporated ("ATSI")

PJM Zone: ATSI

FE Operating Company or
Planning Region:

Ohio Edison

Customer Connection Request

Requested Backfeed Date: N/A

Requested Commercial
Operation Date: 12/31/2021

This study does not imply a FirstEnergy commitment to these dates.

New Facilities

Capacity:	<u>38.69 MW</u>
Energy:	<u>297.66 MW</u>
MFO ¹ :	<u>297.66 MW</u>
Fuel:	<u>Wind</u>

Existing Facilities

Capacity:	<u>N/A</u>
Energy:	<u>N/A</u>
MFO:	<u>N/A</u>
Prior Queue Position(s):	<u>N/A</u>

Point of Interconnection

Primary Point of Interconnection: Beaver-Davis Bessie 345 kV

¹ Maximum Facility Output

Attachment Facilities

There are no Attachment Facilities are required to support this interconnection.

Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Activity Cost	Tax (if applicable)	Total Cost
Construct three (3) breaker 345kV ring bus substation for future four (4) breaker expansion @ AC2-103	\$ 8,890,100	\$ 1,140,600	\$ 10,030,700
Construct a line tap to the proposed AC2-103 switching station. Construct one span from the proposed AC2-103 switching station to the customer POI. @ Beaver-Davis Besse (S-26) 345kV Line Tap to Customer POI	\$ 1,383,500	\$ 177,600	\$ 1,561,100
Project Management, Commissioning, Environmental, Forestry, and SCADA.	\$ 589,700	\$ 75,700	\$ 665,400
Total Direct Connection Facility Costs	\$ 10,863,300	\$ 1,393,900	\$ 12,257,200

Non-Direct Connection Cost Estimate

There are no Non-Direct Connection Facilities are required to support this interconnection.

Description	Activity Cost	Tax (if applicable)	Total Cost
Standard dual SEL421 panel with UPLC for pilot scheme and DTT for the AC2-103 line @ Beaver SS	\$ 285,600	\$ 36,700	\$ 322,300
Standard dual SEL421 panel with UPLC for pilot scheme and DTT for the AC2-103 line @ Davis Bessie SS	\$ 285,600	\$ 36,700	\$ 322,300
Estimated MPLS router at AC2-103 substation to provide SCADA transport for new RTU. Next MPLS hops for SCADA backhaul are assumed to be at Avery substation and Hayes substation.	\$ 186,200	\$ 23,900	\$ 210,100
Total Non-Direct Connection Facility Costs	\$ 757,400	\$ 97,300	\$ 854,700

Transmission Owner Scope of Work

The interconnection of the project will be accomplished by constructing a new 345 kV three (3) breaker ring bus and looping the Beaver-Davis Bessie 345 kV line into the new station. The new substation will be located approximately 28.1 miles from Beaver substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated attachment facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed three breaker ring bus site.

A summary of the connection facilities that will be required for the Primary POI and their estimated costs are shown on the previous page. Based on this scope of work, it is expected to take a minimum of 20 months after the signing of an Interconnection Construction Service Agreement. This includes preliminary payment that compensates FE for the first three months of the engineering design work that is related to the construction of the AC2-103 interconnection substation. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined direct connection and network upgrades, and that PJM will allow all transmission system outages when requested.

Interconnection Customer Requirements

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.
3. The Interconnection Customer seeking to interconnect a wind generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per item 5.iv. of Schedule H to the Interconnection Service Agreement.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

FE Requirements

The Interconnection Customer will be required to comply with all FE Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements

may be found within the “FirstEnergy Requirements for Transmission Connected Facilities” document located at the following links:

<http://www.firstenergycorp.com/feconnect>

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

The IC has requested a non-standard GSU transformer winding configuration. This transformer is in violation of section 14.2.6 of FE’s “Requirements for Transmission Connected Facilities” document and will not be accepted. The GSU transformer must have a grounded wye connection on the high (utility) side and a delta connection on the low (generator) side.

Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 345 kV circuit breaker to protect the AC2-103 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AC2-103 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate

and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

Power Factor Requirements

The IC shall design its wind-powered non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE transmission system.

Network Impacts

The Queue Project AC2-103 was evaluated as a 297.7 MW (Capacity 38.7 MW) injection into the X1-027A 345 kV substation (which is a tap of the Davis Besse – Beaver 345 kV line) in the ATSI area. Project AC2-103 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC2-103 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Contingency Descriptions

The following contingencies resulted in overloads:

None

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Light Load Analysis

Light Load Studies to be conducted during later study phases (applicable to wind, coal, nuclear, and pumped storage projects).

None

Multiple Facility Contingency

(Double Circuit Tower Line contingencies were studied for the full energy output. The contingencies of Line with Failed Breaker and Bus Fault will be performed for the Impact Study.)

None

Affected System Analysis & Mitigation

MISO Impacts:

Results from a recent MISO Affected System Impact Study retool show that there are contingency MISO MTEP upgrades that AC2-103 will need in-service or will need MISO to perform an interim study before going commercial:

MTEP ID	MTEP Cycle	Project Name	Project Description	Expected Completion Date	Status	PJM Conditional Projects
13811	MTEP18	Brownstown - Monroe #1 345 kV Station Equipment Upgrades	Upgrade 345 kV station equipment at Brownstown and Monroe so that the summer emergency rating meets or exceeds 2000 MVA (3347 Amp).	6/1/2020	M3_Long Lead Materials	AC2-103
13813	MTEP18	Monroe - Wayne 345 kV Station Equipment Upgrades	Upgrade 345 kV station equipment at Monroe and Wayne so that the summer emergency rating meets or exceeds 1714 MVA (2868 Am p).	6/1/2020	M3_Long Lead Materials	AC2-103
13814	MTEP18	ITCT-13814- Monroe- Lallendorf_ Upgrades	Upgrade Monroe 345 kV station equipment so that the summer emergency rating meets or exceeds 2000 MVA (3347 Amp).	9/30/2020	M3_Long Lead Materials	AC2-103
13805	MTEP18	Equipment Upgrade at the Belle River - St. Clair 345 kV Station	Upgrade the station equipment at St. Clair to a minimum rating of 2200 MVA (3682 Amp) summer emergency.	6/1/2020	M3_Long Lead Materials	AC2-103

Short Circuit

(Summary of impacted circuit breakers)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

See Attachment 3

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

See Attachment 3

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

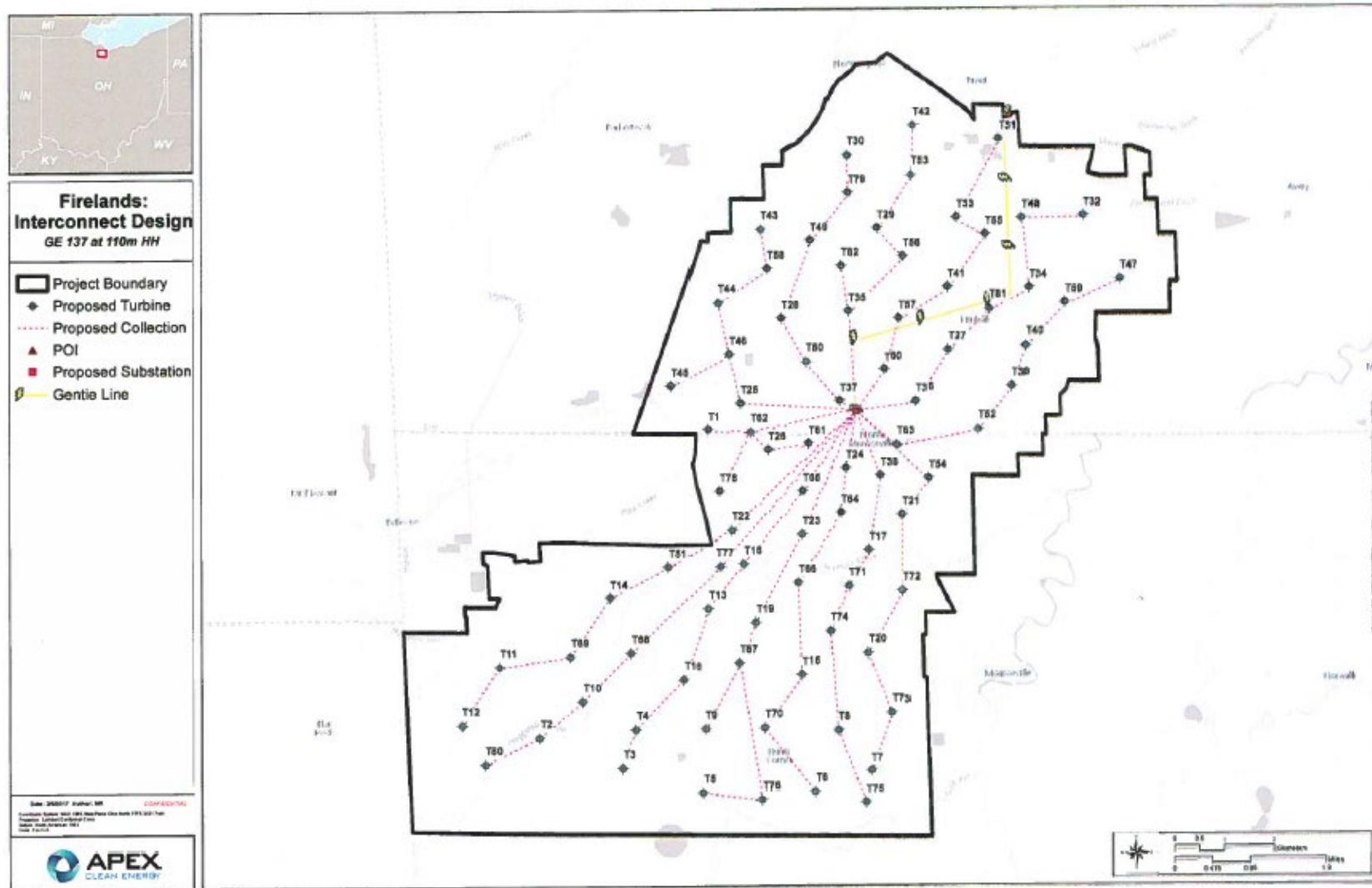
Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

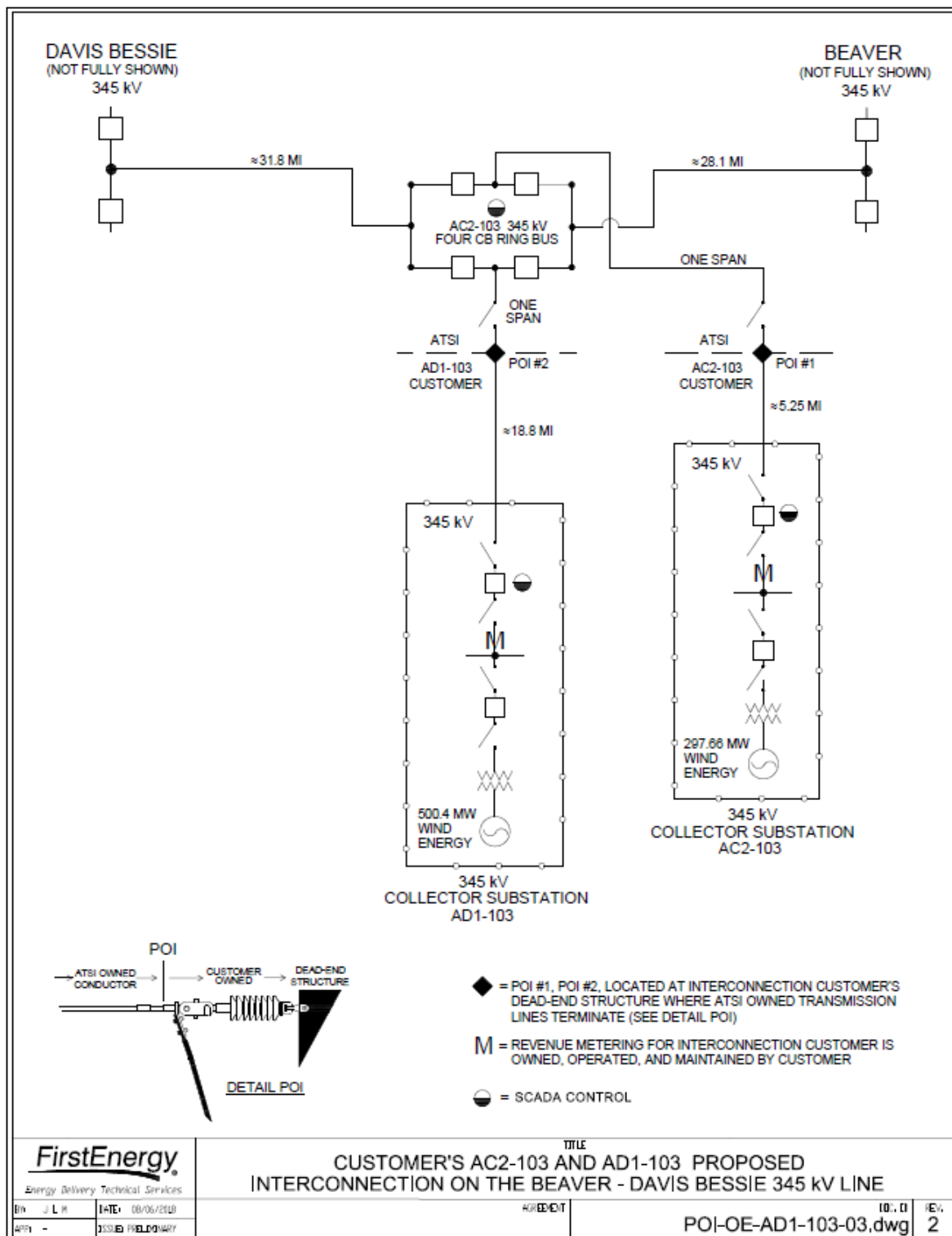
Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

Attachment 1. Project Location



Attachment 2. Single Line Diagram



Attachment 3. Dynamic Simulation Analysis

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Executive Summary

Generator Interconnection Request AC2-103 is for a 297.66 MW Maximum Facility Output (MFO) wind powered generating facility with a Point of Interconnection (POI) on the Davis Besse – Beaver 345 kV circuit in the ATSI system, Huron County, Ohio. AC2-103 and X1-027A would share the same POI switching station.

This report describes a dynamic simulation analysis of AC2-103 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2020 Summer Peak case, modified to include applicable queue projects. AC2-103 has been dispatched online at maximum power output, with POI voltage of (1.014 p.u.), consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 345 kV system.

The AC2-103 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are covered by P1 and P4 contingencies.

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

No relevant High Speed Reclosing (HSR) contingencies were identified.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2020 Summer Peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC2-103 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).

- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Request AC2-103 is for a 297.66 MW Maximum Facility Output (MFO) wind powered generating facility with a Point of Interconnection (POI) on the Davis Besse – Beaver 345 kV circuit in the ATSI system, Huron County, Ohio. AC2-103 and X1-027A would share the same POI switching station.

This analysis is effectively a screening study to determine whether the addition of AC2-103 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AC2-103 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AC2-103 is for a 297.66 MW Maximum Facility Output (MFO) wind powered generating facility with a Point of Interconnection (POI) on the Davis Besse – Beaver 345 kV circuit in the ATSI system, Huron County, Ohio. AC2-103 and X1-027A would share the same POI switching station.

Figure 1 shows the simplified one-line diagram of the AC2-103 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AC2-103 loadflow model.

The dynamic model for the AC2-103 plant is based on the GEWT PSS/E version 33 user defined models.

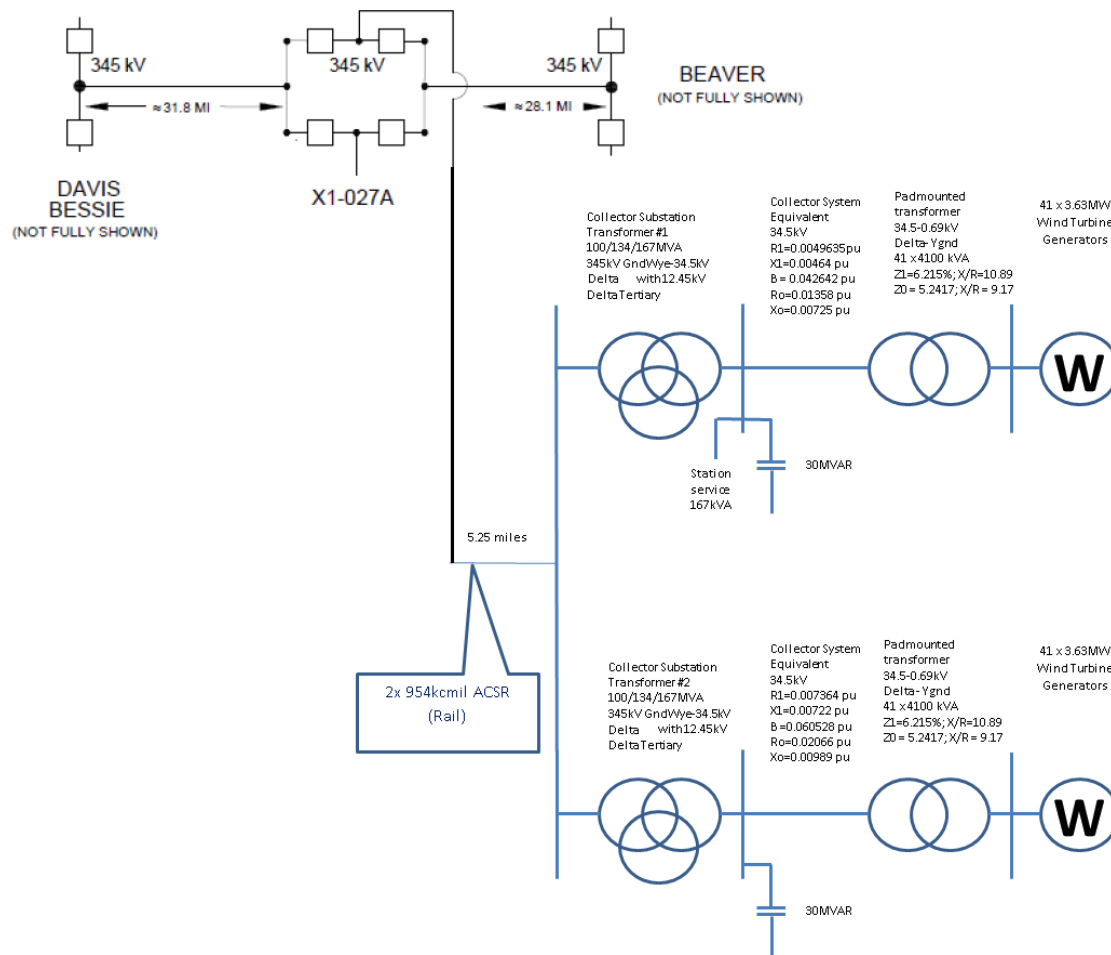


Figure 1: AC2-103 Plant Model

Table 1: AC2-103 Plant Model

	Impact Study Data	Model
Inverters	41 × 3.63 MW GE 3.63 MW wind turbine MVA base = 4.033 MVA Vt = 0.69 kV	Lumped equivalent representing 41 × 3.63 MW GE 3.63 MW wind turbine Pgen 148.83MW Pmax 148.83MW Pmin 0 MW Qmax 58.785MVA Qmin - 58.785MVA Mbase 165.36MVA Zsorce 0.0 + j 0.2 pu @Mbase
	41 × 3.63 MW GE 3.63 MW wind turbine MVA base = 4.033 MVA Vt = 0.69 kV	Lumped equivalent representing 41 × 3.63 MW GE 3.63 MW wind turbine Pgen 148.83MW Pmax 148.83MW Pmin 0 MW Qmax 58.785MVA Qmin - 58.785MVA Mbase 165.36MVA Zsorce 0.0 + j 0.2 pu @Mbase
GSU transformer	41 x 34.5/0.69 kV two winding transformers Transformer base = 4.1 MVA Rating = 4.1 MVA Impedance = 0.00568 + j0.061894 pu @ MVA base Number of taps = 5 Tap step size = 0.025	Lumped equivalent representing 41 x 34.5/0.69 kV two winding transformers Transformer base = 168.1 MVA Rating = 168.1 MVA Impedance = 0.00568 + j0.061894 pu @ MVA base Number of taps = 5 Tap step size = 0.025
	41 x 34.5/0.69 kV two winding transformers Transformer base = 4.1 MVA Rating = 4.1 MVA Impedance = 0.00568 + j0.061894 pu @ MVA base Number of taps = 5 Tap step size = 0.025	Lumped equivalent representing 41 x 34.5/0.69 kV two winding transformers Transformer base = 168.1 MVA Rating = 168.1 MVA Impedance = 0.00568 + j0.061894 pu @ MVA base Number of taps = 5 Tap step size = 0.025

	Impact Study Data	Model
Collector step-up transformer	2 x 345/34.5/12.45 kV three winding transformers Transformer base = 100 MVA Rating = 100 / 134 / 167 MVA Impedances: High – Low = 0.002251 + j 0.088471 pu High – Tertiary = 0.001208 + j 0.046784 pu Low – Tertiary = 0.002994+ j 0.115261 pu All impedances @ MVA base Number of taps = 5 Tap step size = 0.025	2 x 345/34.5/12.45 kV three winding transformers Transformer base = 100 MVA Rating = 100 / 134 / 167 MVA Impedances: High – Low = 0.002251 + j 0.088471 pu High – Tertiary = 0.001208 + j 0.046784 pu Low – Tertiary = 0.002994+ j 0.115261 pu All impedances @ MVA base Number of taps = 5 Tap step size = 0.025
Auxiliary Load	0.1 MW + 0.0 MVAR on LV side of GSU	0.1 MW + 0.0 MVAR on LV side of GSU
Station Load	0.06 MW + 0.00 MVAR on LV side of GSU	0.06 MW + 0.00 MVAR on LV side of GSU
Collector System Equivalent	Positive Sequence Impedance = 0.007364+ j 0.007220 pu Charging Susceptance = j 0.009890pu All impedances @ 100 MVA base	Positive Sequence Impedance = 0.007364+ j 0.007220 pu Charging Susceptance = j 0.009890pu All impedances @ 100 MVA base
Transmission Line	R= 0.000277 X= 0.000872 B= 0.022705	R= 0.000277 X= 0.000872 B= 0.022705

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process².

The selected load flow scenario is the RTEP 2020 Summer Peak case with the following modifications:

- a) Addition of all applicable queue projects prior to AC2-103.
- b) Addition of AC2-103 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC2-103.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AC2-103 initial conditions are listed in Table 2, indicating maximum power output, with AC2-103 regulating POI voltage of (1.014 p.u.), consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 345 kV system.

Table 2: AC2-103 machine initial conditions

Bus	Name	Unit	PGEN (MW)	QGEN (MVAR)	ETERM (p.u.)	POI Voltage (p.u.)
931798	AC2-103 GEN10.6900	1	148.83	20.2	1.05	1.0145
931799	AC2-103 GEN20.6900	1	148.83	20.2	1.05	1.0145

Generation within the vicinity of AC2-103 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AC2-103 is given in Attachment 3.

² Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are covered by P1 and P4 contingencies.

The system was tested for a system intact condition and the fault types listed above. No relevant High Speed Reclosing (HSR) contingencies were studied.

5. Evaluation Criteria

This study is focused on AC2-103, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AC2-103 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%.
- b) The AC2-103 is able to ride through faults (except for faults where protective action trips AC2-103).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

The frequency protection was disabled due to the PSSE deficiency in calculating frequencies for 3ph fault at POIs.

For the fault contingencies tested in this study:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC2-103 generator was able to withstand all contingencies.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Mitigations

No Mitigations were found to be required.

Table 3: Fault list**P0: Steady State**

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.00	3ph @ X1-027A POI – X1-027A Main 345kV line, normal clear loss of X1-027A	5	Stable
P1.01	3ph @ X1-027A POI – AC2-103 345kV line, normal clear loss of AC2-103	5	Stable
P1.02	3ph @ X1-027A POI – Beaver 345kV line, normal clear	5	Stable
P1.03	3ph @ X1-027A POI – Davis Besse 345kV line, normal clear	5	Stable
P1.04	3ph @ Beaver – West Lorain 345kV line, normal clear loss of West Lorain unit 2, 3, 4, 5	5	Stable
P1.05	3ph @ Beaver – Lake Ave 345kV line #1, normal clear	5	Stable
P1.06	3ph @ Beaver – Lake Ave 345kV line #2, normal clear	5	Stable
P1.07	3ph @ Beaver – Carlisle 345kV line, normal clear	5	Stable
P1.08	3ph @ Beaver – X1-027A POI 345kV line, normal clear	5	Stable
P1.09	3ph @ Beaver – Hayes 345kV line, normal clear	5	Stable
P1.10	3ph @ Beaver 345/138kV TF #1, normal clear	5	Stable
P1.11	3ph @ Beaver 345/138kV TF #2, normal clear	5	Stable
P1.12	3ph @ Lake Ave – Beaver 345kV line #1, normal clear	5	Stable
P1.13	3ph @ Lake Ave – Beaver 345kV line #2, normal clear	5	Stable
P1.14	3ph @ Lake Ave – Avon 345kV line #1, normal clear	5	Stable
P1.15	3ph @ Lake Ave – Avon 345kV line #2, normal clear	5	Stable

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.16	3ph @ Lake Ave 345/138kV TF #1, normal clear	5	Stable
P1.17	3ph @ Lake Ave 345/138kV TF #2, normal clear	5	Stable
P1.18	3ph @ Avon – Lake Ave 345kV line #1, normal clear	5	Stable
P1.19	3ph @ Avon – Lake Ave 345kV line #2, normal clear	5	Stable
P1.20	3ph @ Avon 345/138kV TF #91, normal clear	5	Stable
P1.21	3ph @ Avon 345/138kV TF #92, normal clear	5	Stable
P1.22	3ph @ Avon – Juniper 345kV line, normal clear	5	Stable
P1.23	3ph @ Avon unit 9 GSU, normal clear loss of Avon unit 9	5	Stable
P1.24	3ph @ Hayes – Beaver 345kV line, normal clear	5	Stable
P1.25	3ph @ Hayes – Davis Besse 345kV line, normal clear	5	Stable
P1.26	3ph @ Hayes 345/138kV TF, normal clear	5	Stable
P1.27	3ph @ Davis Besse – X1-027A POI 345kV line, normal clear	5	Stable
P1.28	3ph @ Davis Besse – Hayes 345kV line, normal clear	5	Stable
P1.29	3ph @ Davis Besse – Bayshore 345kV line, normal clear	5	Stable
P1.30	3ph @ Davis Besse – Lemoyne 345kV line, normal clear	5	Stable
P1.31	3ph @ Davis Besse unit GSU, normal clear loss of the unit	5	Stable
P1.32	3ph @ Bayshore – Lallendorf 345kV line #1, normal clear	5	Stable
P1.33	3ph @ Bayshore – Lallendorf 345kV line #2, normal clear	5	Stable
P1.34	3ph @ Bayshore – Davis Besse 345kV line, normal clear	5	Stable
P1.35	3ph @ Bayshore 345/138kV TF #1, normal clear	5	Stable
P1.36	3ph @ Bayshore 345/138kV TF #2, normal clear	5	Stable
P1.37	3ph @ Lemoyne – Midway 345kV line, normal clear	5	Stable

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.38	3ph @ Lemoyne – Dowling 345kV line, normal clear	5	Stable
P1.39	3ph @ Lemoyne – Davis Besse 345kV line, normal clear	5	Stable
P1.40	3ph @ Lemoyne – Majestic 345kV line, normal clear	5	Stable
P1.41	3ph @ Lemoyne – Fostoria Central 345kV line, normal clear	5	Stable
P1.42	3ph @ Lemoyne – Troy 345kV line, normal clear loss of Lemoyne units	5	Stable
P1.43	3ph @ Lemoyne 345/138kV TF #1, normal clear	5	Stable
P1.44	3ph @ Lemoyne 345/138kV TF #2, normal clear	5	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.01	SLG @ X1-027A POI 345kV, normal clear loss of X1-027A – Davis Besse 345kV line, SB @ X1-027A POI 345kV, delayed clear loss of X1-027A	5/14	Stable
P4.02	SLG @ X1-027A POI 345kV, normal clear loss of X1-027A – Davis Besse 345kV line, SB @ X1-027A POI 345kV, delayed clear loss of AC2-103	5/14	Stable
P4.03	SLG @ X1-027A POI 345kV, normal clear loss of X1-027A – Beaver 345kV line, SB @ X1-027A POI 345kV, delayed clear loss of X1-027A	5/14	Stable
P4.04	SLG @ X1-027A POI 345kV, normal clear loss of X1-027A – Beaver 345kV line, SB @ X1-027A POI 345kV, delayed clear loss of AC2-103	5/14	Stable
P4.05	SLG @ Beaver 345kV N Bus, normal clear, SB @ Beaver 345kV, delayed clear loss of Beaver 345/138kV TF #1	5/14	Stable
P4.06	SLG @ Beaver 345kV N Bus, normal clear, SB @ Beaver 345kV, delayed clear loss of West Lorain unit 2, 3, 4, 5	5/14	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.07	SLG @ Beaver 345kV N Bus, normal clear, SB @ Beaver 345kV, delayed clear loss of Beaver – Lake Ave 345kV line #1	5/14	Stable
P4.08	SLG @ Beaver 345kV N Bus, normal clear, SB @ Beaver 345kV, delayed clear loss of Beaver – Lake Ave 345kV line #2	5/14	Stable
P4.09	SLG @ Beaver 345kV S Bus, normal clear loss of Beaver 345/138kV TF #2, SB @ Beaver 345kV, delayed clear loss of Beaver – Hayes 345kV line	5/14	Stable
P4.10	SLG @ Beaver 345kV S Bus, normal clear loss of Beaver 345/138kV TF #2, SB @ Beaver 345kV, delayed clear loss of Beaver – X1-027A POI 345kV line	5/14	Stable
P4.11	SLG @ Beaver 345kV S Bus, normal clear loss of Beaver 345/138kV TF #2, SB @ Beaver 345kV, delayed clear loss of Beaver – Carlisle 345kV line	5/65	Stable
P4.12	SLG @ Beaver 345kV S Bus, normal clear loss of Beaver 345/138kV TF #2, SB @ Beaver 345kV, delayed clear loss of Beaver – Lake Ave 345kV line #2	5/14	Stable
P4.13	SLG @ Beaver 345/138kV TF #1, normal clear, SB @ Beaver 345kV, delayed clear loss of Beaver – Hayes 345kV line	5/14	Stable
P4.14	SLG @ Beaver – West Lorain 345kV line, normal clear loss of West Lorain unit 2, 3, 4, 5, SB @ Beaver 345kV, delayed clear loss of Beaver – X1-027A 345kV line	5/14	Stable
P4.15	SLG @ Beaver – Lake Ave 345kV line #1, normal clear, SB @ Beaver 345kV, delayed clear loss of Beaver – Carlisle 345kV line	5/14	Stable
P4.16	SLG @ Lake Ave 345/138kV TF #1, normal clear, SB @ Lake Ave, delayed clear loss of Lake Ave – Beaver 345kV line #1	5/14	Stable
P4.17	SLG @ Lake Ave 345/138kV TF #1, normal clear, SB @ Lake Ave, delayed clear loss of Lake Ave – Beaver 345kV line #2	5/14	Stable
P4.18	SLG @ Lake Ave 345/138kV TF #2, normal clear, SB @ Lake Ave, delayed clear loss of Lake Ave – Avon 345kV line #1	5/14	Stable
P4.19	SLG @ Lake Ave 345/138kV TF #2, normal clear, SB @ Lake Ave, delayed clear loss of Lake Ave – Avon 345kV line #2	5/14	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.20	SLG @ Lake Ave – Avon 345kV line #1, normal clear, SB @ Lake Ave, delayed clear loss of Lake Ave – Beaver 345kV line #1	5/14	Stable
P4.21	SLG @ Lake Ave – Avon 345kV line #2, normal clear, SB @ Lake Ave, delayed clear loss of Lake Ave – Beaver 345kV line #2	5/14	Stable
P4.22	SLG @ Avon – Lake Ave 345kV line #1, normal clear, SB @ Avon, delayed clear loss of Avon unit 9	5/14	Stable
P4.23	SLG @ Avon – Lake Ave 345kV line #1, normal clear, SB @ Avon, delayed clear loss of Avon – Lake Ave 345kV line #2	5/14	Stable
P4.24	SLG @ Avon – Lake Ave 345kV line #2, normal clear, SB @ Avon, delayed clear loss of Avon 345/138kV TF 91	5/14	Stable
P4.25	SLG @ Avon – Juniper 345kV line, normal clear, SB @ Avon, delayed clear loss of Avon 345/138kV TF 91	5/14	Stable
P4.26	SLG @ Avon – Juniper 345kV line, normal clear, SB @ Avon, delayed clear loss of Avon 345/138kV TF 92	5/14	Stable
P4.27	SLG @ Avon 345/138kV TF 92, normal clear, SB @ Avon, delayed clear loss of Avon unit 9	5/14	Stable
P4.28	SLG @ Carlisle – North Medina 345kV line, normal clear, SB @ Carlisle, delayed clear loss of Carlisle 345kV bus	5/14	Stable
P4.29	SLG @ Hayes 345kV, normal clear loss of Hayes 345/138kV TF, SB @ Hayes 345kV, delayed clear loss of Hayes – Beaver 345kV line	5/14	Stable
P4.30	SLG @ Hayes 345kV, normal clear loss of Hayes 345/138kV TF, SB @ Hayes 345kV, delayed clear loss of Hayes – Davis Besse 345kV line	5/14	Stable
P4.31	SLG @ Hayes 345kV, normal clear loss of Hayes – Davis Besse 345kV line, SB @ Hayes 345kV, delayed clear loss of Hayes – Beaver 345kV line	5/14	Stable
P4.32	SLG @ Hayes 345kV, normal clear loss of Hayes – Beaver 345kV, SB @ Hayes 345kV, delayed clear loss of line Hayes – Davis Besse 345kV line	5/14	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.33	SLG @ Davis Besse 345kV bus 1, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse – Hayes 345kV line	5/14	Stable
P4.34	SLG @ Davis Besse 345kV bus 1, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse – Lemoyne 345kV line	5/14	Stable
P4.35	SLG @ Davis Besse 345kV bus 1, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse unit	5/14	Stable
P4.36	SLG @ Davis Besse 345kV bus 2, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse – Hayes 345kV line	5/14	Stable
P4.37	SLG @ Davis Besse 345kV bus 2, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse – Bayshore 345kV line	5/14	Stable
P4.38	SLG @ Davis Besse 345kV bus 2, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse – X1-027A POI 345kV line	5/14	Stable
P4.39	SLG @ Davis Besse – Bayshore 345kV line, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse unit	5/14	Stable
P4.40	SLG @ Davis Besse – Lemoyne 345kV line, normal clear, SB @ Davis Besse, delayed clear loss of Davis Besse – X1-027A POI 345kV line	5/14	Stable
P4.41	SLG @ Bayshore 345kV, normal clear loss of Bayshore 345/138kV TF #1, SB @ Bayshore, delayed clear loss of Bayshore – Lallendorf 345kV line #1	5/14	Stable
P4.42	SLG @ Bayshore 345kV, normal clear loss of Bayshore 345/138kV TF #1, SB @ Bayshore, delayed clear loss of Bayshore – Lallendorf 345kV line #2	5/14	Stable
P4.43	SLG @ Bayshore 345kV, normal clear loss of Bayshore 345/138kV TF #2, SB @ Bayshore, delayed clear loss of Bayshore – Lallendorf 345kV line #1	5/14	Stable
P4.44	SLG @ Bayshore 345kV, normal clear loss of Bayshore 345/138kV TF #2, SB @ Bayshore, delayed clear loss of Bayshore – Davis Besse 345kV line	5/14	Stable
P4.45	SLG @ Bayshore 345kV, normal clear loss of Bayshore – Lallendorf 345kV line #2, SB @ Bayshore, delayed clear loss of Bayshore – Davis Besse 345kV line	5/14	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.46	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne 345/138kV TF #1, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne – Davis Besse 345kV line	5/14	Stable
P4.47	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne 345/138kV TF #1, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne – Majestic 345kV line	5/14	Stable
P4.48	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne 345/138kV TF #2, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne – Davis Besse 345kV line	5/14	Stable
P4.49	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne 345/138kV TF #2, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne – Midway 345kV line	5/14	Stable
P4.50	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne – Midway 345kV line, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne – Dowling 345kV line	5/14	Stable
P4.51	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne – Dowling 345kV line, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne units	5/14	Stable
P4.52	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne – Fostoria Central 345kV line, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne units	5/14	Stable
P4.53	SLG @ Lemoyne 345kV, normal clear loss of Lemoyne – Fostoria Central 345kV line, SB @ Lemoyne 345kV, delayed clear loss of Lemoyne – Majestic 345kV line	5/14	Stable

P5: SLG Fault with Delayed (Zone 2) Clearing

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P5.01	SLG @ 80% of X1-027A POI – Beaver 345kV line, relay failure @ X1-027A POI, delayed clear	5/65	Stable
P5.02	SLG @ 80% of X1-027A POI – Davis Besse 345kV line, relay failure @ X1-027A POI, delayed clear	5/65	Stable
P5.03	SLG @ 80% of Beaver – X1-027A POI 345kV line, relay failure @ Beaver, delayed clear	5/65	Stable
P5.04	SLG @ 80% of Beaver – Hayes 345kV line, relay failure @ Beaver, delayed clear	5/65	Stable
P5.05	SLG @ 80% of Beaver – Lake Ave 345kV line #1, relay failure @ Beaver, delayed clear	5/65	Stable
P5.06	SLG @ 80% of Beaver – Lake Ave 345kV line #2, relay failure @ Beaver, delayed clear	5/65	Stable

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P5.07	SLG @ 80% of Beaver – Carlisle 345kV line, relay failure @ Beaver, delayed clear	5/65	Stable
P5.08	SLG @ 80% of Davis Besse – X1-027A POI 345kV line, relay failure @ Davis Besse, delayed clear	5/65	Stable
P5.09	SLG @ 80% of Davis Besse – Hayes 345kV line, relay failure @ Davis Besse, delayed clear	5/65	Stable
P5.10	SLG @ 80% of Davis Besse – Bayshore 345kV line, relay failure @ Davis Besse, delayed clear	5/65	Stable
P5.11	SLG @ 80% of Davis Besse – Lemoyne 345kV line, relay failure @ Davis Besse, delayed clear	5/65	Stable

P7: Tower Failure

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P7.01	CONTINGENCY 'C5-CEI-345-001' /* AVON-Lake Ave #1 AND #2 345KV LINE OUTAGES	5	Stable
P7.02	CONTINGENCY 'C5-CEI-345-002' /* AVON-JUNIPER AND AVON-Lake Ave #1 345KV LINE OUTAGES	5	Stable
P7.03	CONTINGENCY 'C5-CEI-345-010' /* AVON-JUNIPER AND HARDING-CHAMBERLIN 345KV LINE OUTAGES	5	Stable
P7.04	CONTINGENCY 'C5-CEI-345-011' /* AVON-JUNIPER AND EMILY-FOX 138KV LINE OUTAGES	5	Stable
P7.05	CONTINGENCY 'C5-OEC-138-015' /* BEAV-CARL, BEAV-DAVISBESSE 345	5	Stable
P7.06	CONTINGENCY 'C5-OEC-345-001' /* BEAVER-LAKAVE 345 CK 1 & 2	5	Stable
P7.07	CONTINGENCY 'C5-TE-138-021' /* BAYSH - Y1-069 1 & 2 345	5	Stable
P7.08	CONTINGENCY 'C5-TE-138-022' /* Y1-069-MONROE/LEMOYNE-MAJ 345	5	Stable
P7.09	CONTINGENCY 'C5-TE-138-023' /* Y1-069-FOSTORIA/LEMOYNE-MAJ 345	5	Stable
P7.10	CONTINGENCY 'C5-TE-138-024' /* Y1-069-FOSTORIA/LEMOYNE-FOSTORIA 345	5	Stable

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P7.11	Beaver – Hayes 345kV line Beaver – X1-027A POI 345kV line	5	Stable
P7.12	Beaver – Hayes 345kV line Davis Besse – X1-027A POI 345kV line	5	Stable
P7.13	Davis Besse – Hayes 345kV line Davis Besse – X1-027A POI 345kV line	5	Stable

Attachment 1. PSS/E Model One Line Diagram

Attachment 2. PSS/E Dynamic Model

/******

/*AC2-103

/*GE 3.63MW X 82 in total

/*Modeled as two Main TFs, each takes 41*3.63MW

/******

931798 'USRMDL' 1 'GEWTG2' 1 1 4 18 3 5

0 41 0 0

3.6300 0.80000 0.50000 0.90000 1.2200 1.2000
2.0000 0.40000 0.80000 10.000 0.20000E-01 0.0000
0.0000 0.50000 0.16700 0.90000 0.92500 0.0000 /

931798 'USRMDL' 1 'GEWTE2' 4 0 12 67 18 9

931798 0 0 1 0 0

0 0 0 1 0 0
0.50000 2.000 1.0000 0.0000 0.0000 0.50000E-01 3.0000
0.60000 1.1200 0.40000E-01 0.43600 -0.43600 1.1000 0.20000E-01
0.45000 -0.45000 60.000 0.41000 0.90000
1.1000 40.000 0.50000 1.4500 0.50000E-01
0.50000E-01 1.0000 0.15000 0.96000 0.99600
1.0040 1.0400 1.00000 1.0000 1.00000
0.40000 1.0000 0.20000 1.0000 0.25000
-1.0000 14.0000 25.000 3.0000 -0.90000
8.0000 0.2000 10.000 1.0000 1.7000
1.22 1.2500 5.0000 0.0000 0.0000
0.000 0.25000E-02 1.0000 5.5000 0.10000
-1.0000 0.10000 0.0000 0.10000 -0.10000
0.70000 0.12000 -0.12000 /

931798 'USRMDL' 1 'GEWTT1' 5 0 1 5 4 3 0

2.0400 0.0000 0.0000 1.8800 1.5000 /

931798 'USRMDL' 1 'GEWGD1' 505 0 1 6 0 4 0

9999.0 5.0000 30.000 9999.0 9999.0
30.000 /

931798 'USRMDL' 1 'GEWTA2' 505 0 0 9 1 4

20.000 0.0000 27.000 -4.0000 0.0000 1.2250
67.000 121.80 1200.0 /

931798 'USRMDL' 0 'GEWTP2' 505 0 1 10 3 3 0

0.30000 150.00 25.000 3.0000 30.000
-4.0000 27.000 -10.000 10.000 1.0000 /

93179800 'USRMSC' 'GEWPLT2' 512 0 2 0 0 17 931798 '1' /

93179801 'VTGTPAT'

931798 931798 '1'

0.10000 5.0000 0.50000 0.80000E-01/

93179802 'VTGTPAT'

931798 931798 '1'

0.20000 5.0000 0.693179800 0.80000E-01/

93179803 'VTGTPAT'

931798 931798 '1'

0.40000 5.0000 1.00000 0.80000E-01/

93179804 'VTGTPAT'
 931798 931798 '1 '
 0.60000 5.0000 1.70000 0.80000E-01/
 93179805 'VTGTPAT'
 931798 931798 '1 '
 0.75000 5.0000 2.20000 0.80000E-01/
 93179806 'VTGTPAT'
 931798 931798 '1 '
 0.85000 5.0000 10.0000 0.80000E-01/
 93179807 'VTGTPAT'
 931798 931798 '1 '
 0.90000 5.0000 600.0000 0.80000E-01/
 93179808 'VTGTPAT'
 931798 931798 '1 '
 0.00000 1.1010 1.000 0.80000E-01/
 93179809 'VTGTPAT'
 931798 931798 '1 '
 0.00000 1.1500 0.5000 0.80000E-01/
 93179810 'VTGTPAT'
 931798 931798 '1 '
 0.00000 1.17500 0.2000 0.80000E-01/
 93179811 'VTGTPAT'
 931798 931798 '1 '
 0.00000 1.200 0.1000 0.80000E-01/
 93179812 'VTGTPAT'
 931798 931798 '1 '
 0.00000 1.3000 0.0100 0.80000E-01/
 93179813 'FRQTPAT' 931798 931798 '1' 55.0 65.0 0.10 0.08 /
 93179814 'FRQTPAT' 931798 931798 '1' 57.0 63.0 60.0 0.08 /

931799 'USRMDL' 1 'GEWTG2' 1 1 4 18 3 5
 0 41 0 0
 3.6300 0.80000 0.50000 0.90000 1.2200 1.2000
 2.0000 0.40000 0.80000 10.000 0.20000E-01 0.0000
 0.0000 0.50000 0.16700 0.90000 0.92500 0.0000 /
 931799 'USRMDL' 1 'GEWTE2' 4 0 12 67 18 9
 931799 0 0 1 0 0
 0 0 0 1 0 0
 0.50000 2.000 1.0000 0.0000 0.0000 0.50000E-01 3.0000
 0.60000 1.1200 0.40000E-01 0.43600 -0.43600 1.1000 0.20000E-01
 0.45000 -0.45000 60.000 0.41000 0.90000
 1.1000 40.000 0.50000 1.4500 0.50000E-01
 0.50000E-01 1.0000 0.15000 0.96000 0.99600
 1.0040 1.0400 1.00000 1.0000 1.00000
 0.40000 1.0000 0.20000 1.0000 0.25000
 -1.0000 14.0000 25.000 3.0000 -0.90000
 8.0000 0.2000 10.000 1.0000 1.7000
 1.22 1.2500 5.0000 0.0000 0.0000
 0.000 0.25000E-02 1.0000 5.5000 0.10000
 -1.0000 0.10000 0.0000 0.10000 -0.10000

	0.70000	0.12000	-0.12000	/					
931799 'USRMDL' 1 'GEWTT1'			5	0	1	5	4	3	0
	2.0400	0.0000	0.0000	1.8800		1.5000	/		
931799 'USRMDL' 1 'GEWGD1'			505	0	1	6	0	4	0
	9999.0	5.0000	30.000	9999.0		9999.0			
	30.000	/							
931799 'USRMDL' 1 'GEWTA2'			505	0	0	9	1	4	
	20.000	0.0000	27.000	-4.0000		0.0000	1.2250		
	67.000	121.80	1200.0	/					
931799 'USRMDL' 0 'GEWTP2'			505	0	1	10	3	3	0
	0.30000	150.00	25.000	3.0000		30.000			
	-4.0000	27.000	-10.000	10.000		1.0000	/		
93179900 'USRMSC' 'GEWPLT2' 512	0	2	0	0	17				
						931799	'1'	/	

93179901 'VTGTPAT'				
931799	931799	'1'		
0.10000	5.0000	0.50000	0.80000E-01/	
93179902 'VTGTPAT'				
931799	931799	'1'		
0.20000	5.0000	0.693179900	0.80000E-01/	
93179903 'VTGTPAT'				
931799	931799	'1'		
0.40000	5.0000	1.00000	0.80000E-01/	
93179904 'VTGTPAT'				
931799	931799	'1'		
0.60000	5.0000	1.70000	0.80000E-01/	
93179905 'VTGTPAT'				
931799	931799	'1'		
0.75000	5.0000	2.20000	0.80000E-01/	
93179906 'VTGTPAT'				
931799	931799	'1'		
0.85000	5.0000	10.0000	0.80000E-01/	
93179907 'VTGTPAT'				
931799	931799	'1'		
0.90000	5.0000	600.0000	0.80000E-01/	
93179908 'VTGTPAT'				
931799	931799	'1'		
0.00000	1.1010	1.000	0.80000E-01/	
93179909 'VTGTPAT'				
931799	931799	'1'		
0.00000	1.1500	0.5000	0.80000E-01/	
93179910 'VTGTPAT'				
931799	931799	'1'		
0.00000	1.17500	0.2000	0.80000E-01/	
93179911 'VTGTPAT'				
931799	931799	'1'		
0.00000	1.200	0.1000	0.80000E-01/	
93179912 'VTGTPAT'				
931799	931799	'1'		
0.00000	1.3000	0.0100	0.80000E-01/	
93179813 'FRQTPAT'	931798	931798	'1'	55.0 65.0 0.10 0.08 /

93179814 'FRQTPAT' 931798 931798 '1' 57.0 63.0 60.0 0.08 /

Attachment 3. PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
238554	02AVONG7 13.800	7	1	94.6	94.6	0	0	40.26	0
238555	02AVONG9 20.000	9	1	672.1	672.1	0	49.96	308.5	-263
238564	02BAYSG1 18.000	1	1	136	136	0	44.76	62.33	-29.8
238571	02BEAVGA 13.800	A	1	51.6	51.6	10	-6	23.91	-6
238572	02BEAVGB 13.800	B	1	53.6	53.6	10	-6	22.2	-6
238601	02FRMENG 1 18.000	1	1	184.7	184.7	0	-1.2	121.7	-88.7
238602	02FRMENG 2 18.000	2	1	184.7	184.7	0	-1.2	121.7	-88.7
238603	02FRMENG 3 23.000	3	1	344	344	0	-1.2	233.3	-173
238670	02DVBSG1 25.000	1	1	960	960	0	203.9	478.8	52.6
238683	02EASTG5 24.000	S5	1	0	0	0	20.98	519	-150
238885	02LEMOG1 18.000	1	1	160	160	90	30.01	100	-45
238886	02LEMOG2 18.000	2	1	160	160	90	30.01	100	-45
238887	02LEMOG3 18.000	3	1	160	160	90	30.01	100	-45
238888	02LEMOG4 18.000	4	1	160	160	90	30.01	100	-45
238965	02MNF DG1 18.000	1	1	906	906	400	147.1	260	-435
238966	02MNF DG2 18.000	2	1	907	907	400	147.1	250	-435
238967	02MNF DG3 18.000	3	1	892	892	400	147.1	270	-435
238979	02NAPMUN 138.00	1	1	25	25	24	0.327	14	-10
238979	02NAPMUN 138.00	2	1	12	12	10	0.157	5	-6
239035	02PERRG1 22.000	1	1	1331	1331	1294	-21.6	600	-270
239087	02SAMMG3 20.000	3	1	206.6	206.6	55	68.25	114.7	-39.3
239088	02SAMMG4 20.000	4	1	201.1	201.1	77	68.25	116.5	-40.9
239089	02SAMMG5 24.000	5	1	329.4	329.4	135	68.25	183.1	-70.5
239090	02SAMMG6 20.000	6	1	666.4	666.4	250	68.25	391.1	-185
239091	02SAMMG7 20.000	7	1	673.4	673.4	250	68.25	359.3	-210
239171	02WLORG-2 13.800	2	1	82.6	82.6	45	-26.5	63.94	-26.5
239172	02WLORG-3 13.800	3	1	84.9	84.9	45	-26.6	64.34	-26.6
239173	02WLORG-4 13.800	4	1	84.7	84.7	45	-26.2	62.87	-26.2
239174	02WLORG-5 13.800	5	1	85	85	45	-26.4	63.71	-26.4
239175	02WLORG-6 13.800	6	1	85	85	45	-26.3	63.14	-26.3
239267	02AVG10 13.800	10	1	21	21	0	5	5	5
239293	02BS-PKR 4.1600	99	1	14.6	14.6	0	1.64	8	-4
240950	02BG5 69.000	1	1	11	11	0	0.24	20	0
240950	02BG5 69.000	2	1	11	11	0	0.24	12.7	0
241908	02LLF_W4-0044.1600	1	1	19.2	19.2	0	0	0	0
247270	05RPMNG1 18.000	1	1	151.1	151.1	75	17.61	88.48	-57.3

247271	05RPMNG2 18.000	2	1	151	151	75	17.61	105.1	-56.2
247272	05RPMNG3 18.000	3	1	151	151	75	17.61	84.87	-56.8
247549	V3-028 C 345.00	1C	1	7.6	7.6	0	0	0	0
247932	U2-072_GEN 0.6600	1	1	300	300	0	0	0	0
247940	U4-028 0.6900	1	1	100.8	100.8	0	0	0	0
247941	U4-029 0.6900	1	1	100.8	100.8	0	0	0	0
247948	V3-028 E 345.00	1E	1	12.4	12.4	0	0	0	0
253900	15BVRVL1 22.000	1	1	999	999	998	215.6	412	37.8
253901	15BVRVL2 22.000	2	1	999	999	999	215.6	411	38.2
256338	18PALISD 22.000	1	1	837	837	150	65.38	349.2	-312
264500	19COLFX 41.570	11	1	12.1	12.1	0	0	0	0
264759	19MON12P 345.00	11	1	12.05	12.05	0	0	0	0
264854	19ENFPP 22.000	1	1	1146	1146	1102	430	430	-40
264871	19MON1 26.000	1	1	752.3	752.3	350	261.5	327	-184
264872	19MON2 26.000	2	1	741.9	741.9	370	261.5	452	-175
264873	19MON3 26.000	3	1	782.2	782.2	325	243.5	492	-150
264874	19MON4 26.000	4	1	762	762	350	243.5	441	-164
907210	X1-027A WT1 0.6000	1	1	126	126	1	0	0	0
907211	X1-027A WT2 0.6000	1	1	126	126	1	0	0	0
907212	X1-027A WT3 0.6000	1	1	126	126	1	0	0	0
907213	X1-027A WT4 0.6000	1	1	126	126	1	0	0	0
913441	AA1-056_ST1 19.000	1	1	336.2	336.2	0	25.57	220	-163
913442	AA1-056_CT2 19.000	1	1	324.3	324.3	0	33.49	202	-151
913443	AA1-056_CT1 19.000	1	1	324.3	324.3	0	33.49	202	-151
918001	AA1-006 69.000	1	1	0.4	0.4	0	0.192	0.192	-0.13
929595	U2-041 GEN 0.6900	1	1	151	151	0	43.39	49.4	-49.4
929596	U2-041 GEN 0.6900	1	1	149	149	0	43.39	48.6	-48.6
931798	AC2-103 GEN10.6900	1	1	148.8	148.8	0	22.75	58.79	-58.8
931799	AC2-103 GEN20.6900	1	1	148.8	148.8	0	22.75	58.79	-58.8